VICTORY 2027

ELECTRICITY AND MAGNETISM

LECTURE-02 PHYSICS

Shailendra sir



Topics To Be Covered

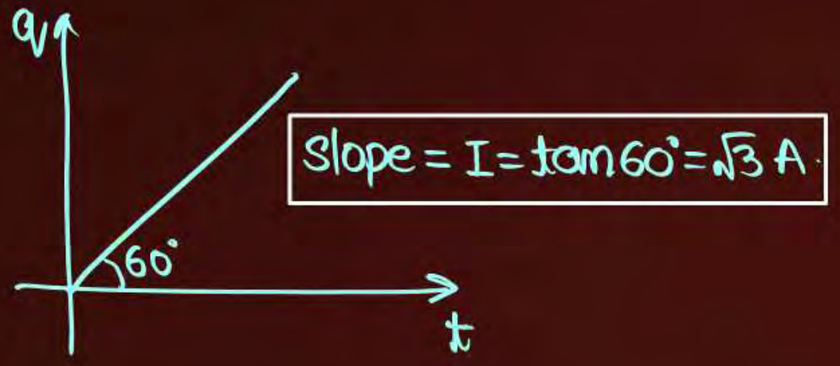
1. REVISION
2. OHM'S LAW
3. E.M.F. OF A CELL
4. QUESTIONS


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Physics Wallah

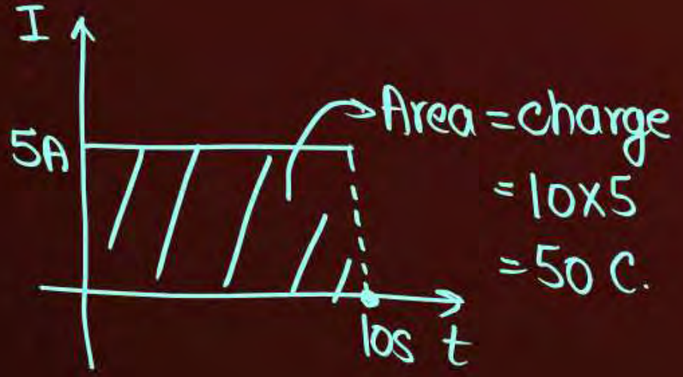


Revision

$$I = \frac{\Delta q}{\Delta t}$$



Slope = $I = \tan 60° = \sqrt{3}$ A



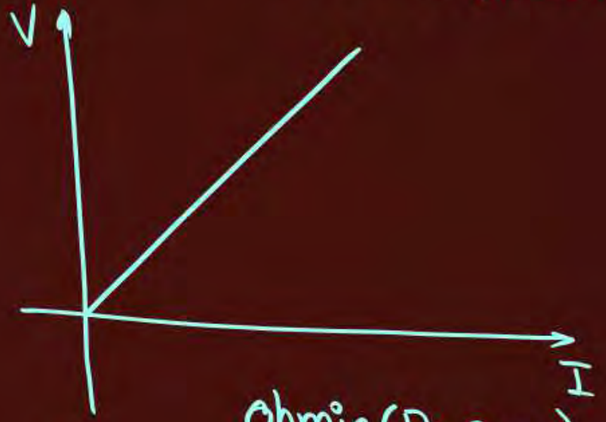
Ohm's Law

$V \propto I$

$V = IR$

Constant (Resistance)

V → Potential Difference across conductor.



Ohmic (R=Cons.)



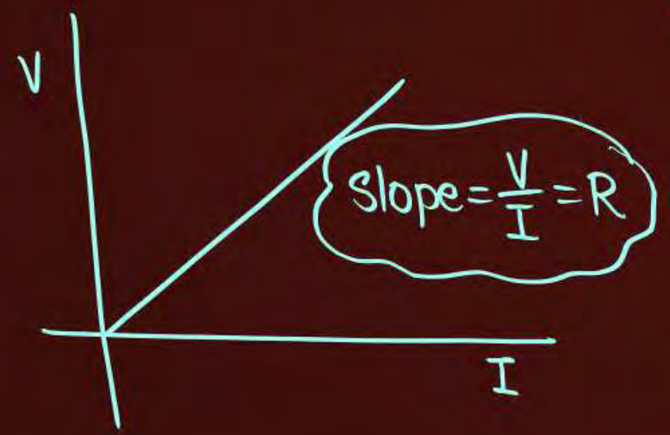
Non-ohmic
R ≠ Const.

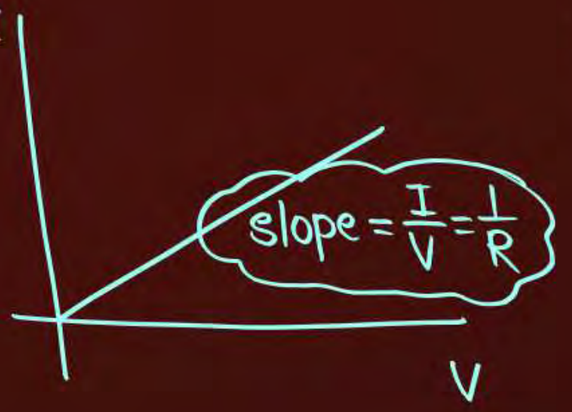
V-I GRAPH FOR A CONDUCTOR AT TWO TEMPERATURES

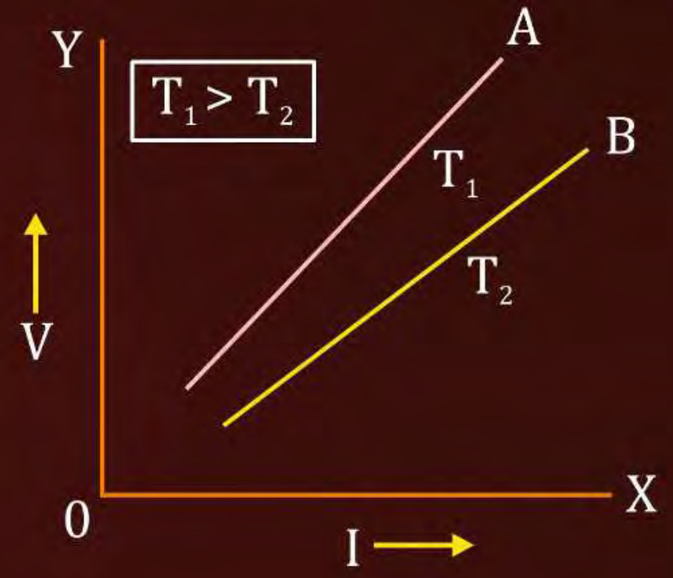
Conductor $T \uparrow$ collision $\uparrow$ $R \uparrow$

$T_1 > T_2$

$\boxed{R_1 > R_2}$







QUESTIONS

For which of the following substances, resistance decreases with increase in temperature?

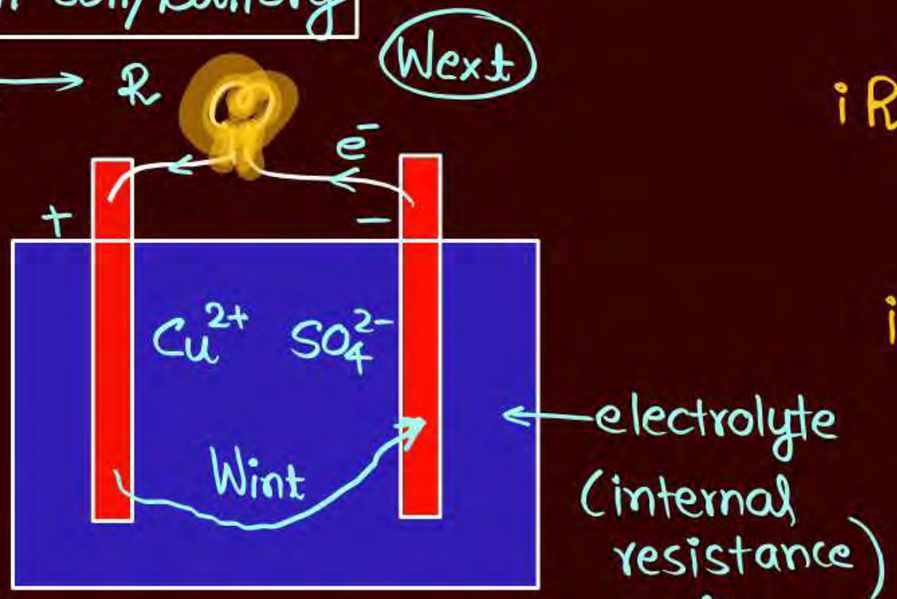
$T \uparrow R \downarrow$ (Semiconductor)

Si, Ge

1. copper
2. mercury
3. carbon
4. platinum



PhD on Cell/Battery



$$iR = \frac{W_{ext}}{q} = V \quad \text{(Terminal Pot. Difference)}$$

$$ir = \frac{W_{int}}{q} = v \quad \text{(Potential drop across internal resistance)}$$

$$\frac{W_{total}}{q} = \mathcal{E} \quad \text{(electromotive force of cell)}$$

$$\frac{W_{total}}{q} = \frac{W_{ext}}{q} + \frac{W_{int.}}{q}$$

$$\mathcal{E} = V + v$$

$$\boxed{V = \mathcal{E} - v}$$

$$iR = \mathcal{E} - ir$$

$$\text{Terminal Pot. Diff.} = \text{EMF} - \text{Pot. Drop.}$$

$\mathcal{E} > V$ (Discharging)

$\mathcal{E} < V$ (charging)

$V = IR$

$V = Ir$

$V = \mathcal{E} - ir$

if $i = 0$

$V = \mathcal{E}$



E.M.F. OF A CELL

The e.m.f. of a cell is defined as the **energy spent (or the work done) per unit charge in taking a positive test charge around the complete circuit of the cell** (i.e., in the circuit outside the cell as well as in the electrolyte inside the cell).

When no current is drawn from a cell i.e., when the cell is in open circuit, the **potential difference between the terminals of the cell** is called its electro-motive force (or e.m.f.).

If W work is done in taking a test charge q around the complete circuit of the cell, then e.m.f. of the cell is

$$\varepsilon = \frac{W}{q}$$



FACTORS AFFECTING THE E.M.F. OF A CELL

The e.m.f. of a cell depends on:
(1) the **material of the electrodes**, and
(2) the **electrolyte** used in the cell.

However, it is independent of
(a) the shape of electrodes,
(b) the distance between the electrodes, and
(c) the amount of electrolyte.

Thus the **e.m.f. of a cell is its characteristic property**. It is different for different kind of cells.

- For example, the e.m.f. of a voltaic cell is 1.08 volt, that of a Leclanche cell is 1.5 volt and of a Daniel cell is 1.08 volt.



TERMINAL VOLTAGE OF A CELL

When current is drawn from a cell i.e., when the cell is in closed circuit, the **potential difference between the electrodes of the cell** is known as its terminal voltage.

The terminal voltage of a cell is defined as the **work done per unit charge in carrying a positive test charge around the circuit connected across the terminals of cell**. The terminal voltage of a cell is generally denoted by the letter V. It is also expressed in volt (symbol V).

If W′ is the work done in carrying a test charge q around the circuit connected across the terminals of a cell (i.e., only outside the cell and not inside the cell), then the terminal voltage of the cell is

$$\mathbf{V = \frac{W'}{q}}$$

VOLTAGE DROP IN A CELL

If w is the work done (or energy spent) in carrying a test charge q through the electrolyte inside the cell, then the quantity $\frac{w}{q}$ is called the voltage drop in the cell which is denoted as v,

Voltage drop in the cell

$$v = \frac{w}{q}$$

Thus, the work done in carrying a unit charge through the electrolyte is called the voltage drop in the cell. It is so called because this voltage is not available to us for use.

Relationship between e.m.f. and terminal voltage of a cell

By the law of conservation of energy,

$$\mathbf{W = W' + w}$$

Dividing the above eqn. (i) by q on both the sides, we get

$$\frac{W}{q} = \frac{W'}{q} + \frac{w}{q}$$

Therefore

$$\varepsilon = V + v$$

$$\boxed{\mathbf{V = \varepsilon - v}}$$

Thus when current is drawn from a cell, its terminal voltage V is less than its e.m.f. ε by an amount equal to the voltage drop v inside the cell.



Distinction between e.m.f. and terminal voltage of a cell

	E.m.f. of cell		Terminal voltage of cell
1.	It is measured by the amount of work done in moving a unit positive charge in the complete circuit inside and outside the cell	1.	It is measured by the amount of work done in moving a unit positive charge in the circuit outside the cell.
2.	It is the characteristic of the cell, i.e., it does not depend on the amount of current drawn from the cell.	2.	It depends on the amount of current drawn from the cell. More the current drawn from the cell, less is the terminal voltage.
3.	It is equal to the terminal voltage when cell is not in use, while greater than the terminal voltage when cell is in use.	3.	It is equal to the e.m.f. of cell when cell is not in sue, while less than the e.m.f. when cell is in use.

INTERNAL RESISTANCE OF A CELL

→ electrolyte

The **resistance offered by the electrolyte inside the cell**, to the flow of current, is called the internal resistance of cell. It is the internal resistance of cell due to which its voltage drops on drawing current from it. It is denoted by the symbol r. Its unit is ohm (symbol Ω).

If current I is drawn from the cell of which internal resistance is r, the voltage drop is

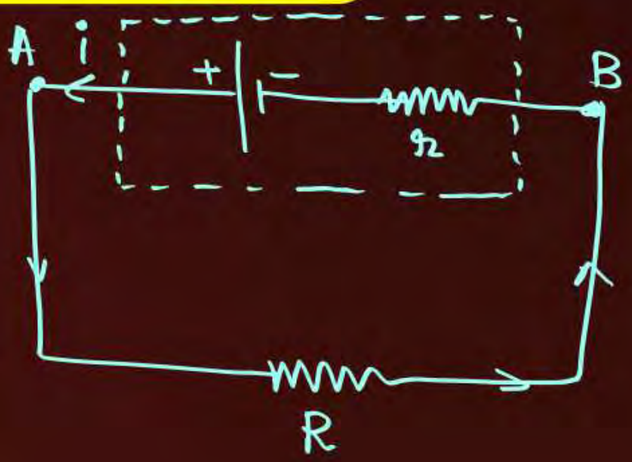
$$\boxed{v = Ir}$$



Relationship between the e.m.f., terminal voltage and internal resistance

$$V = \mathcal{E} - ir \quad \text{——} ①$$

$$i = \frac{\mathcal{E}}{R+r} \quad \text{——} ②$$



from eq ① & ②

$$V = \mathcal{E} - \left(\frac{\mathcal{E}}{R+r}\right) r$$

$$V = \frac{\mathcal{E}(R+r) - \mathcal{E}r}{R+r}$$

$$V(R+r) = \mathcal{E}R + \mathcal{E}r - \mathcal{E}r$$

$$VR + Vr = \mathcal{E}R$$

$$Vr = \mathcal{E}R - VR \Rightarrow \boxed{r = \frac{(\mathcal{E}-V)R}{V}}$$

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↓ ↓ ↓

Total Memory | Usable Memory | Software

↓ ↓ ↓

E | V | v

FACTORS AFFECTING THE INTERNAL RESISTANCE OF A CELL

(1) **The surface area of the electrodes** - larger the surface area of electrodes, less is the internal resistance. $r \propto \frac{1}{A}$

(2) **The distance between the electrodes** - more the distance between the electrodes, greater is the internal resistance. $r \propto d$

(3) **The nature and concentration of the electrolyte** - less ionic the electrolyte or higher the concentration of electrolyte, greater is the internal resistance. $r \propto C$

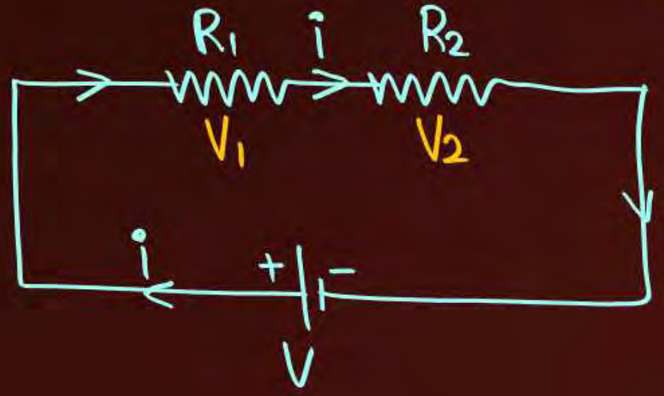
(4) **The temperature of the electrolyte** - higher the temperature of the electrolyte, less is the internal resistance. $T \uparrow r \downarrow$

COMBINATION OF RESISTORS

① SERIES COMBINATION

Same currents

different Voltage



$V = V_1 + V_2$

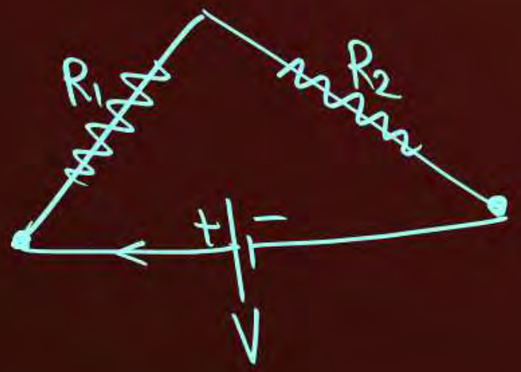
$iR = iR_1 + iR_2$

$$R = R_1 + R_2$$

equivalent resistance.

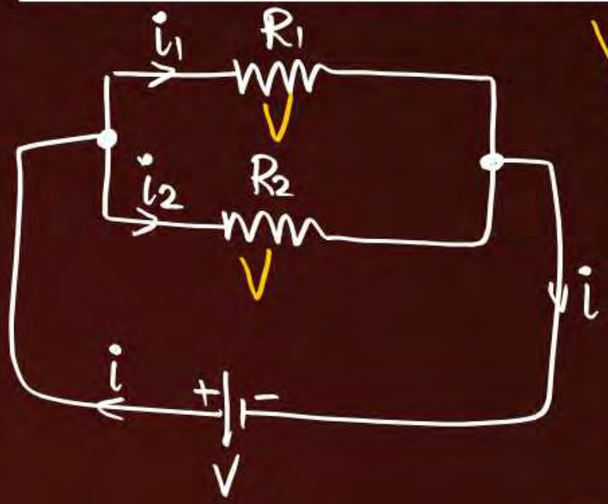
↓

Maximum



② PARALLEL COMBINATION

same voltage different current.



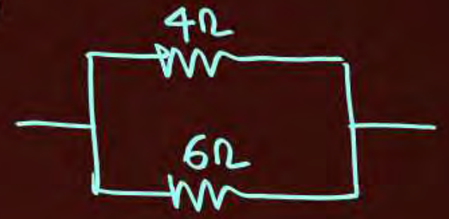
$i = i_1 + i_2$

$\frac{V}{R} = \frac{V}{R_1} + \frac{V}{R_2}$

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$$

↓

minimum



$\frac{1}{R} = \frac{1}{4} + \frac{1}{6}$

$\frac{1}{R} = \frac{3+2}{12}$

$R = \frac{12}{5}\Omega$

$$R = 2.4\Omega$$

R↑

@ Parallel

(b) Series

QUESTIONS

A high resistance voltmeter measures the potential difference across a battery to be 9.0 V. On connecting a 24Ω resistor across the terminals of the battery, the voltmeter reads 7.2V. Calculate the internal resistance of the battery.

1. 2
2. 4
3. 6
4. 8

$\mathcal{E} = 9V$

$R = 24\Omega$

$V = 7.2$ volts

$V = iR$

$7.2 = i(24)$

$i = \frac{7.2}{24}$ A

$V = \mathcal{E} - ir$

$7.2 = 9 - \frac{7.2}{24} \times r$

$r = \left(\frac{\mathcal{E}-V}{V}\right) R$

$= \frac{9-7.2}{7.2} \times 24$

QUESTIONS

A battery of voltage 12 V sends current 400 mA to an appliance. After a long continuous use, the current drops to 320 mA and the appliance does not operate.

Find:

(i) the resistance of appliance, $V = iR \Rightarrow R = \frac{V}{i} = \frac{12}{400 \times 10^{-3}} = \frac{12000}{400} = 30\,\Omega$

(ii) the terminal voltage of the battery when the appliance stops operating, $V = iR$

$= 320 \times 10^{-3} \times 30$

$= 9.6$ V.

(iii) the voltage drop, and $v = \mathcal{E} - V = 12 - 9.6 = 2.4$ volts.

(iv) the internal resistance of the battery.

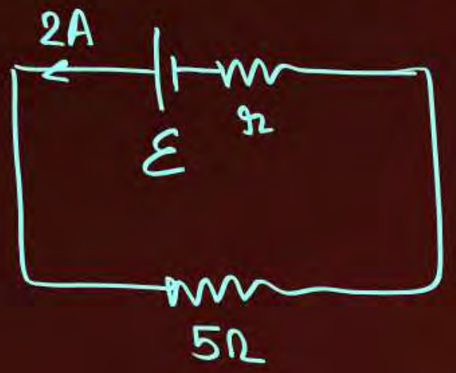
$v = ir$

$2.4 = 400 \times 10^{-3}\, r$

$r = \frac{2.4 \times 1000}{400} = 6\,\Omega$

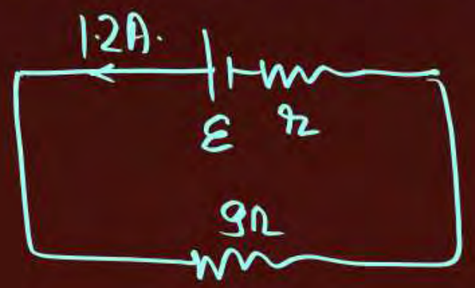
QUESTIONS

A cell supplies a current of 2 A when it is connected to a 5Ω resistance and supplies a current of 1.2 A, if it is connected to a resistance of 9Ω. Find the e.m.f and internal resistance of the cell.



$$i = \frac{\varepsilon}{R+r}$$

$$2 = \frac{\varepsilon}{5+r} \quad \text{—①}$$



$$1.2 = \frac{\varepsilon}{9+r} \quad \text{—②}$$

eq ① / eq ②

$$\frac{2}{1.2} = \frac{9+r}{5+r}$$

$$10+2r = 10.8+1.2r$$

$$0.8r = 0.8 \Rightarrow \boxed{r = 1\Omega}$$

$$\boxed{\varepsilon = 12V}$$

QUESTIONS

What resistance must be connected to a 15Ω resistance to provide an effective resistance of 6Ω?

1. 5Ω
2. 10Ω
3. 15Ω
4. 20Ω

$$\frac{1}{6} = \frac{1}{15} + \frac{1}{R}$$

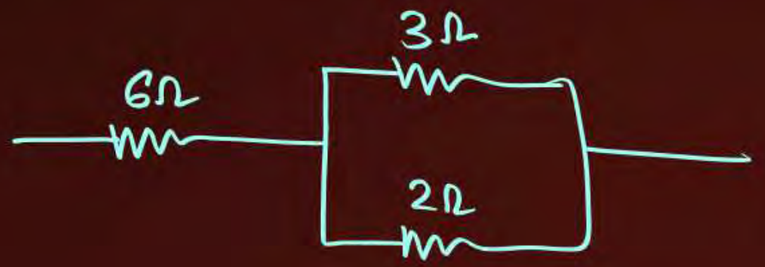
$$\frac{1}{R} = \frac{1}{6} - \frac{1}{15}$$

$$\frac{1}{R} = \frac{5-2}{30}$$

$$R = 10\Omega$$

QUESTIONS

Three resistors of $6\Omega, 3\Omega$ and 2Ω are connected together so that the total resistance is greater than 6Ω but less than 8Ω. Draw a diagram to show this arrangement and calculate its total resistance.



$$R_{eq} = 6 + \frac{3\times2}{3+2} = 6 + \frac{6}{5} = \frac{36}{5}\,\Omega.$$

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$\frac{1}{R} = \frac{R_2 + R_1}{R_1 R_2}$$

$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$$

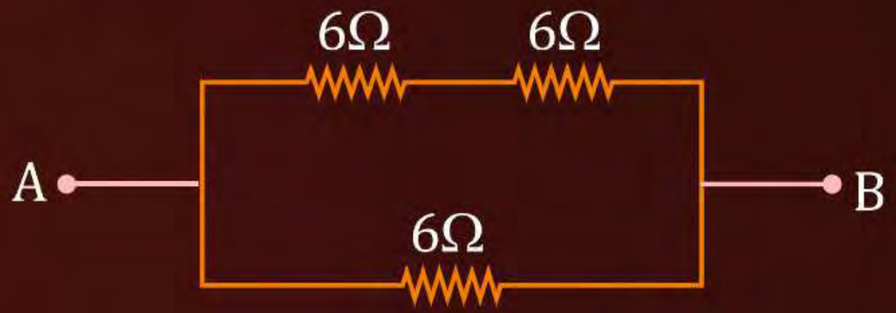
$$= \frac{12 \times 6}{12 + 6} = \frac{72}{18} = 4\,\Omega$$

QUESTIONS

Calculate the equivalent resistance between the points A and B in the circuit shown.

1. 3Ω
2. 4Ω
3. 5Ω
4. 6Ω

$$R = \frac{12 \times 6}{12+6} = \frac{72}{18} = 4\Omega$$

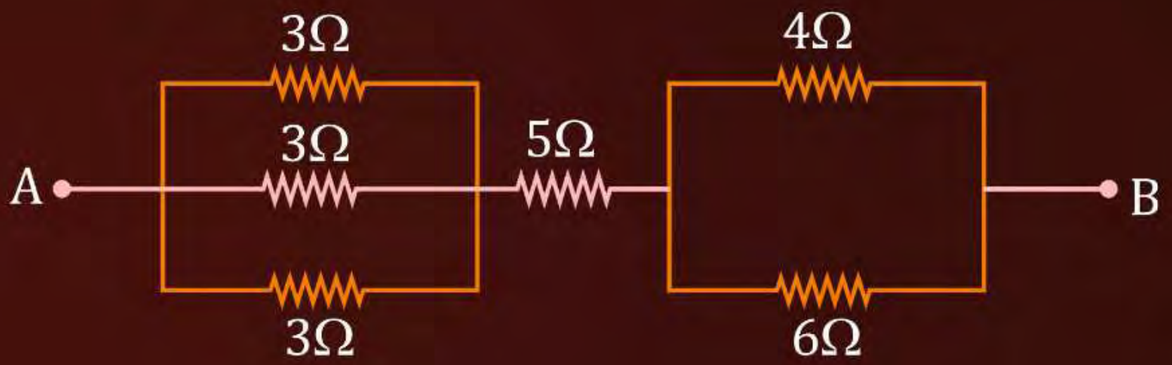


QUESTIONS

HW

Calculate the equivalent resistance between the points A and B in the circuit shown.

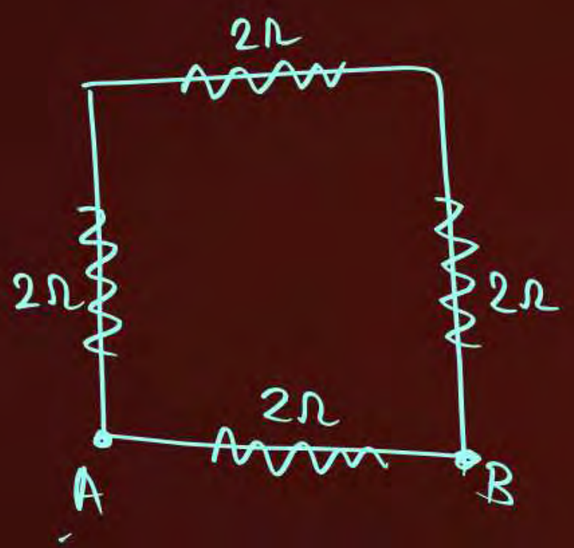
1. 6.4Ω
2. 8.4Ω
3. 10.4Ω
4. 12.4Ω



QUESTIONS

Four resistors each of resistance 2.0 Ω are joined end to end to form a square ABCD. Calculate the equivalent resistance of the combination between any two adjacent corners.

$\frac{12}{8}\Omega$



QUESTIONS

HW

For the combination of resistors shown, find the equivalent resistance between the point

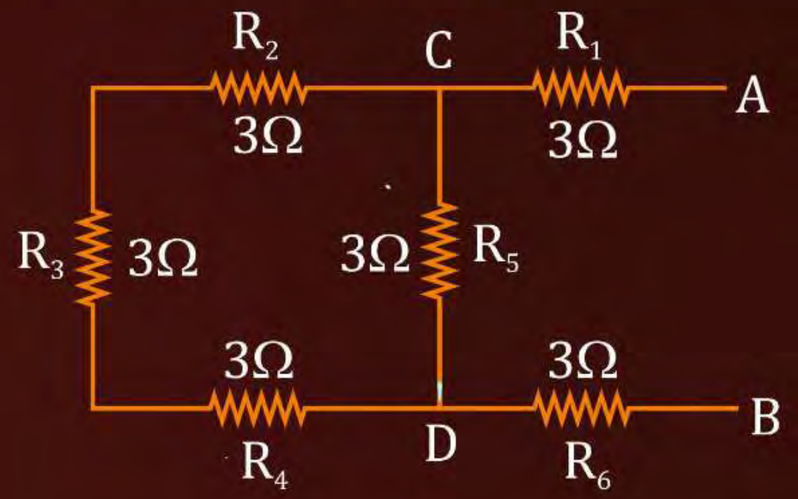
(a) C and D, and

(b) A and B.

$$R_{AB} = 6 + \frac{27}{12}$$

$$= \frac{72+27}{12}$$

$$= \frac{99}{12}\ \Omega.$$



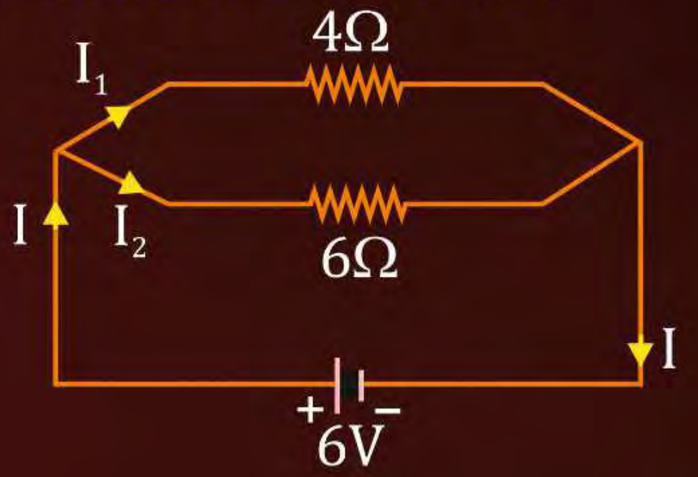
QUESTIONS

Two resistors of 4Ω and 6Ω are connected in parallel. The combination is connected across a 6 V battery of negligible resistance. Calculate:

(a) the total resistance of circuit,

(b) the current through the battery,

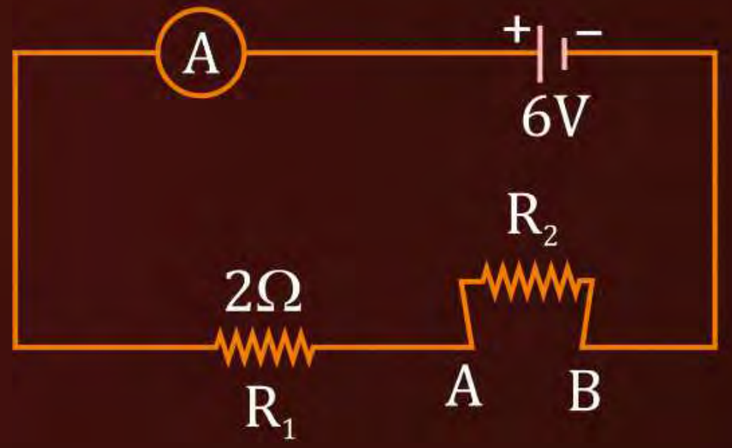
(c) the current through each resistor. Draw a circuit diagram.



QUESTIONS

The circuit diagram includes a 6 V battery, an ammeter A, a fixed resistor R_1 of 2Ω and a resistance wire R_2 connected between the terminals A and B. The resistance of the battery and ammeter may be neglected. Calculate the ammeter readings when the wire R_2 is of

(a) 0.20 m length and of resistance 4Ω.
(b) 0.40 m length and of same thickness and material as in case (a).
(c) 0.20 m length and of area of cross section double than in case (a).



HOMEWORK

- REVISE THIS LECTURE AGAIN.
- PRACTISE 20 QUESTIONS DAILY.
- COMMENT YOUR RESPONSE ABOUT THIS CLASS.

Thank You